Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD):

FR-SQG-001: Generate SQL Query for Sales Orders

• Title: Generate SQL Query for Sales Orders

• Description: The system must generate a complete SQL SELECT statement that applies transformation logic, includes appropriate joins, and ensures correct data types and formatting based on the provided tabular mapping document.

• Preconditions:

• The tabular mapping document is available with the required columns (Column Name, Column Type, Source Table, Source Column/Transformation Logic, Join Condition).

• The input data is available in the specified source tables.

• Main Flow / Functional Steps:

1. **Create Intermediate Table "sales_orders_comp_stg"**:

• Retrieve the mapping document for "sales_orders_comp_stg".

• Generate a SQL SELECT statement that applies transformation logic and joins as specified in the mapping document.

• Use aliases (a, b, etc.) consistently as per the join logic.

• Ensure correct data types and formatting (e.g., date conversions).

2. **Create Main Table "sales_orders_comp"**:

• Retrieve the mapping document for "sales_orders_comp".

• Generate a SQL SELECT statement that applies transformation logic and joins as specified in the mapping document.

• Use aliases (a, b, etc.) consistently as per the join logic.

• Ensure correct data types and formatting (e.g., date conversions).

3. **Apply Transformation Logic and Joins**:

• For each column in the target table, apply the specified transformation logic or retrieve the data from the source table(s) as specified.

• Perform joins as specified in the mapping document, using the correct join conditions and aliases.

4. **Handle Conditional Logic**:

• Apply conditional logic (e.g., IF, CASE) as described in the mapping document.

5. **Format the SQL Query**:

• Format the generated SQL query for readability, using indentation and line breaks as necessary.

• Detailed Requirements:

• The system must generate a SQL query that includes all necessary joins and filters (e.g., DATA_OPERATION <> 'D').

• The system must use aliases consistently (a, b, etc.) as per the join logic.

• The system must ensure that all transformation logic is correctly applied.

• The system must handle conditional logic (e.g., IF, CASE) as described in the mapping document.

• Specific Requirements for "sales_orders_comp_stg":

• The system must concatenate "referenceto" and "refno" to generate "CompOrderNumber".

• The system must perform a left join between "b_source.ord_hdr" and "b_source.ord_dtl" on " referenceto" and "refno".

• The system must apply the specified transformation logic for "CompCreateDate" and " CompPromisedDeliveryDate".

• Specific Requirements for "sales_orders_comp":

• The system must retrieve data from "s_master.sale_orders" and apply transformation logic as specified.

• The system must perform joins with other tables (e.g., "b_source.ord_addr", " b_source.ord_channel") as specified in the mapping document.

• The system must apply conditional logic (e.g., IF) as described in the mapping document for columns like "CompSalesDistributionChannel".